NONCOOPERATIVE GAME THEORY — ECE270: HOMEWORK #6

Exercise 13 (Sudoku)**.** Consider the multi-player version of the Sudoku game discussed in Section 13.5, but with the outcome of player P_i given by

$$J_i(\gamma) := \sigma^{\text{row}}_{r_i}(\gamma) + \sigma^{\text{col}}_{c_i}(\gamma) + \sigma^{\text{block}}_{b_i}(\gamma), \qquad \gamma := \{\gamma_1, \gamma_2, \ldots, \gamma_N\},$$

where $r_i \in \{1,2,\ldots,9\}$, $c_i \in \{1,2,\ldots,9\}$, and $b_i \in \{1,2,\ldots,9\}$ denote player P_i's row, column, and block, respectively; and $\sigma^{\text{row}}_r(\gamma)$ denotes the total number of times that a digit is repeated in the rth row; $\sigma^{\text{col}}_c(\gamma)$ denotes the number of times that a digit is repeated in the cth column; and $\sigma^{\text{block}}_b(\gamma)$ the number of times that a digit is repeated in the bth block.

Show that this defines an exact potential game potential with potential

$$\phi(\gamma) := \frac{1}{9}\sum_{i=1}^{N} J_i(\gamma) = \sum_{r=1}^{9} \sigma^{\text{row}}_r(\gamma) + \sum_{c=1}^{9} \sigma^{\text{col}}_c(\gamma) + \sum_{b=1}^{9} \sigma^{\text{block}}_b(\gamma). \qquad \square$$

Exercise 14 (Sudoku)**.** Write a MATLAB® script to solve the Sudoku puzzle in Figure 13.2 by computing improvement paths for the multi-player version of the Sudoku game considered in in Section 13.5, but with the outcomes in Exercise 13.

Plot an histogram with the values of the potential obtained for 1,000 runs of your algorithm. For each run, randomize both the initial actions of the players and the order you select players for the generation of the improvement policy, to get different Nash equilibria over the 1,000 run. This type of randomization is done in the code in the solution to Practice Exercise 13.5.

Hint: Make use of the code in MATLAB® Hint 4. You will likely get an histogram like the one in Figure 2, which was obtained with 10,000 runs. Note that the number of times that we obtained a zero potential—the global minimum—is relatively small (about 1.0%*).* □

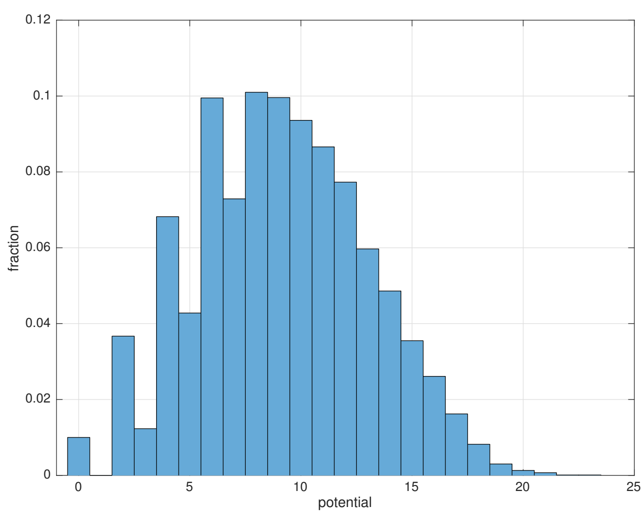

Figure 2. Histogram of the values of the potential at the Nash equilibria obtained by computing 10,000 improvements paths for the multi-player version of the Sudoku game in Exercise 14.